

Unit 6: Cubic & Cube Root Functions

Test Review: Transformations & Inverses – ANSWER KEY

Name: TEACHER KEY

Period: _____

Part 1: Quick Concepts

1. A function and its inverse are reflections across the line $y = x$.
 2. If $f(7) = 20$, then $f^{-1}(20) =$ 7.
 3. If $f(-5) = 12$, what point is on $f^{-1}(x)$? $(12, -5)$
 4. If $f(6) = -3$, what point is on $f(x)$? $(6, -3)$
-

Part 2: Identify Parameters & Describe Transformations

5. $f(x) = -3(x + 2)^3 - 4$

$$a = \underline{-3} \quad h = \underline{-2} \quad k = \underline{-4}$$

Transformations: **Reflect x-axis, Vert Stretch 3, Left 2, Down 4**

6. $f(x) = \frac{1}{2}\sqrt[3]{x - 5} + 3$

$$a = \underline{1/2} \quad h = \underline{5} \quad k = \underline{3}$$

Transformations: **Vert Compress 1/2, Right 5, Up 3**

Part 3: Write Equations from Descriptions

7. $f(x) = x^3$ is reflected over the x-axis, shifted left 4, and up 6.

$$g(x) = \underline{-(x + 4)^3 + 6}$$

8. $f(x) = \sqrt[3]{x}$ is vertically stretched by 2, shifted right 3, and down 1.

$$g(x) = \underline{2\sqrt[3]{x - 3} - 1}$$

9. The function $g(x) = (x - c)^3 + d$ has its center point at $(-3, 7)$.

$$c = \underline{3} \quad d = \underline{7}$$

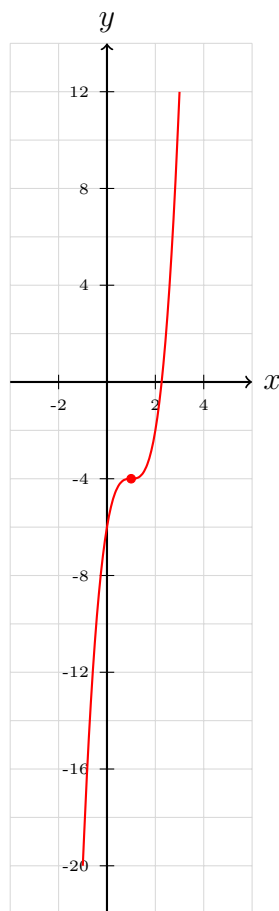
Part 4: Graph Transformations

10. $f(x) = 2(x - 1)^3 - 4$

$a = \underline{2} \quad h = \underline{1} \quad k = \underline{-4}$

Center: $(\underline{1}, \underline{-4})$

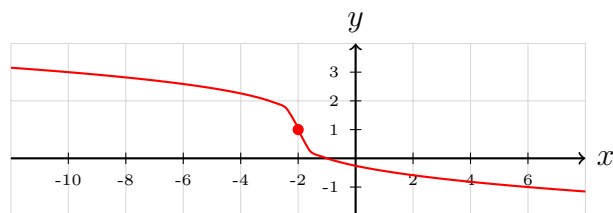
Parent	New
$(-2, -8)$	$(\underline{-1}, \underline{-20})$
$(-1, -1)$	$(\underline{0}, \underline{-6})$
$(0, 0)$	$(\underline{1}, \underline{-4})$
$(1, 1)$	$(\underline{2}, \underline{-2})$
$(2, 8)$	$(\underline{3}, \underline{12})$

y-int: $\underline{-6}$ x-int: $\underline{2.26}$ 

11. $f(x) = -\sqrt[3]{x + 2} + 1$
 $a = \underline{-1} \quad h = \underline{-2} \quad k = \underline{1}$

Center: $(\underline{-2}, \underline{1})$

Parent	New
$(-8, -2)$	$(\underline{-10}, \underline{3})$
$(-1, -1)$	$(\underline{-3}, \underline{2})$
$(0, 0)$	$(\underline{-2}, \underline{1})$
$(1, 1)$	$(\underline{-1}, \underline{0})$
$(8, 2)$	$(\underline{6}, \underline{-1})$

y-int: $\underline{-0.26}$ x-int: $\underline{-1}$ 

Part 5: Transform Specific Points

12. $g(x) = -3(x - 2)^3 + 5.$

$$(1, 1) \longrightarrow (\mathbf{3} , \mathbf{2}) \qquad (-2, -8) \longrightarrow (\mathbf{0} , \mathbf{29})$$

Find the x-intercept of $g(x)$: ≈ 3.19

13. $g(x) = -\sqrt[3]{x - 4} + 2.$

$$(8, 2) \longrightarrow (\mathbf{12} , \mathbf{0}) \qquad (-1, -1) \longrightarrow (\mathbf{3} , \mathbf{3})$$

Find the x-intercept of $g(x)$: $\mathbf{12}$ **Part 6: Write Equations from Graphs**

14. (Graph omitted for brevity)

Center: $(\mathbf{2}, -\mathbf{2})$

$$f(x) = \underline{\underline{\frac{1}{2}(x - 2)^3 - 2}}$$

15. (Graph omitted for brevity)

Center: $(-\mathbf{2}, \mathbf{1})$

$$f(x) = \underline{\underline{-\sqrt[3]{x + 2} + 1}}$$

Part 7: Find Inverse Functions16. Find the inverse of $f(x) = (x + 4)^3 - 2$

$$y = (x + 4)^3 - 2 \rightarrow x = (y + 4)^3 - 2$$

$$x + 2 = (y + 4)^3 \rightarrow \sqrt[3]{x + 2} = y + 4$$

$$f^{-1}(x) = \underline{\underline{\sqrt[3]{x + 2} - 4}}$$

17. Find the inverse of $f(x) = (2x - 1)^3 + 3$

$$x = (2y - 1)^3 + 3 \rightarrow x - 3 = (2y - 1)^3$$

$$\sqrt[3]{x - 3} = 2y - 1 \rightarrow \sqrt[3]{x - 3} + 1 = 2y$$

$$f^{-1}(x) = \underline{\underline{\frac{\sqrt[3]{x - 3} + 1}{2}}}$$

Part 8: Error Analysis

18. Marcus was asked to shift $f(x) = x^3$ up 4 units and right 2 units. He wrote $g(x) = (x + 2)^3 + 4$.
 Explain: **He used +2 inside (Left shift) instead of -2 (Right shift).**

Correct equation: $g(x) = \underline{\underline{(x - 2)^3 + 4}}$

19. Destiny says $f(x) = 3(x - 1)^3 + 2$ is vertically compressed. Find and correct her error:
Since a = 3 and $|3| > 1$, it is a Vertical STRETCH, not a compression.

Part 9: Multiple Choice**20. D****21. A****22.** Vertical stretch: 4 Vertical shift: Up 5 Horizontal shift: Right 2**Part 10: Intercepts****23.** $g(x) = 2(x + 1)^3 - 16$ **(a)** y-intercept: $2(1)^3 - 16 = -14$ Ans: -14**(b)** x-intercept: $2(x + 1)^3 = 16 \rightarrow (x + 1)^3 = 8 \rightarrow x + 1 = 2 \rightarrow x = 1$ Ans: 1**Part 11: Verify****24.** Verify $f(f^{-1}(x)) = x$:

$$\begin{aligned}
 f(f^{-1}(x)) &= ((\sqrt[3]{x-2} + 1) - 1)^3 + 2 \\
 &= (\sqrt[3]{x-2})^3 + 2 \\
 &= x - 2 + 2 \\
 &= \mathbf{x} \quad \checkmark
 \end{aligned}$$